

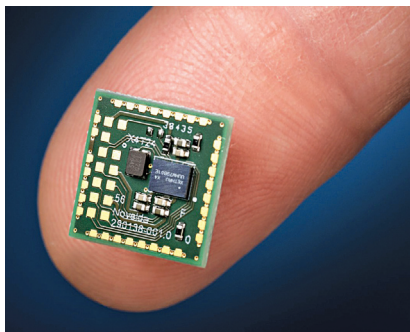
TRULY INNOVATIVE TECH

INNOVATION UPDATES

Amongst numerous press releases of new products received by us, these are the ones we found worthy of the title *Truly Innovative Tech*

Human-presence sensor

The latest NOVELDA ultra-sideband (UWB) sensors for smart home and building automation give any device the ability to accurately sense human



presence. Groundbreaking UWB impulse radar can detect human presence and motion, providing more accurate touch-free interaction with everyday products, including laptops, smart refrigerators, and lighting. The sensor works by detecting tiny movements made when we breathe to perceive human presence. The sensor is so robust it can sense human presence even if subjects are lying under a duvet or wearing several layers of clothing. Causing no impact on product design, it can be completely hidden in a device and can operate through glass, plastic, or wood.

NOVELDA
www.novelda.com

Smallest UWB module

The world's smallest ultra-wideband (UWB) module combines elevated levels of accuracy and reliability with low power to ensure compact and battery-driven IoT devices operate as efficiently and cost-effectively as possible. Using short-range radio frequency (RF) technology, the Type 2AB UWB + Bluetooth low-energy (BLE) connectivity module can be

used in a wide variety of applications where precise detection is essential, including social distancing to minimise the spread of Covid-19. Ideal for

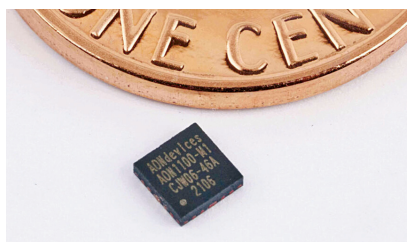


manufacturing engineers developing advanced products for the IoT market, the system-in-package module incorporates three antennas—two UWB for phased difference of arrival functionality and one for BLE.

Murata Manufacturing
www.murata.com

Motion-sensing AI edge chip

The new AON1100 edge AI processor enables unprecedented sensing capabilities in battery-powered always-on devices that need to support local



wake words, voice commands, sound event detection, context detection, and sensor fusion at high accuracy—all while maintaining industry-leading low power consumption that extends battery life, which results in a better end-user experience. It also simultaneously detects specific motion patterns, such as walking or falling. When used in phones, headsets, wearables, game controllers, vehicles, or smart home

appliances, the AON1100 enables natural human-machine interface at the device level without sending any data to the cloud, improving the user experience and guaranteeing privacy.

AONDevices
www.aondevices.com

Long-range sub-GHz SoCs

The new sub-1GHz (sub-GHz) SoCs deliver the world's first sub-GHz wireless solutions that combine long-range RF and energy efficiency with certified ARM PSA Level 3 security to meet the



global demand for high-performance, battery-powered IoT products. The EFR32FG23 (FG23) and EFR32ZG23 (ZG23) system-on-chip (SoC) solutions provide developers with flexible, multi-protocol sub-GHz connectivity options, supporting a wide range of modulation schemes and advanced wireless technologies, including Amazon Sidewalk, mioty, Wireless M-Bus, Z-Wave, and proprietary IoT networks.

Silicon Labs
www.silabs.com

Solid-state ToF lidar

2D/3D dual solid-state ToF lidar helps translate the 3D world and a 3D object into a 2D image using precisely measured depth data, so it can be used for security, industrial automation, drones, touch-screen, and more. The budget-friendly device weighs only 28gm and